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%% Battery Design Metrics and Chemistry Visualization Script
% This script generates professional plots for lab verification data
% and compares NMC vs LFP battery chemistry characteristics.

clear; clc; close all;

%% =====
%% FIGURE 1: OPERATIONAL VERDICT & LAB VALUES
%% =====
figure('Name', 'Design Verification: Target vs Lab', 'Color', 'w',
'Position', [100, 100, 900, 500]);
tiledlayout(1, 2, 'TileSpacing', 'Compact', 'Padding', 'Compact');

% --- Subplot 1: Voltage Metrics ---
nexttile;
voltage_labels = {'Low UV Trip', 'Open Circuit V', 'High OV Trip'};
% Targets (using midpoints for ranges where applicable)
voltage_targets = [60.0, 72.2, 84.0];
voltage_measured = [60.05, 72.42, 83.92];

b1 = bar([voltage_targets; voltage_measured]', 'grouped');
b1(1).FaceColor = '#0072BD'; % Target Blue
b1(2).FaceColor = '#D95319'; % Measured Orange
grid on;
xticklabels(voltage_labels);
ylabel('Voltage (V)', 'FontSize', 11, 'FontWeight', 'bold');
title('Voltage Threshold Verification', 'FontSize', 12, 'FontWeight',
'bold');
legend({'Target / Limit', 'Measured Lab Value'}, 'Location', 'northwest');

% Add pass/fail annotations directly onto bars
text(1, voltage_measured(1)+1, 'Pass', 'HorizontalAlignment', 'center',
'FontWeight', 'bold', 'Color', 'g');
text(2, voltage_measured(2)+1, 'Pass', 'HorizontalAlignment', 'center',
'FontWeight', 'bold', 'Color', 'g');
text(3, voltage_measured(3)+1, 'Pass', 'HorizontalAlignment', 'center',
'FontWeight', 'bold', 'Color', 'g');
ylim([0, 100]);

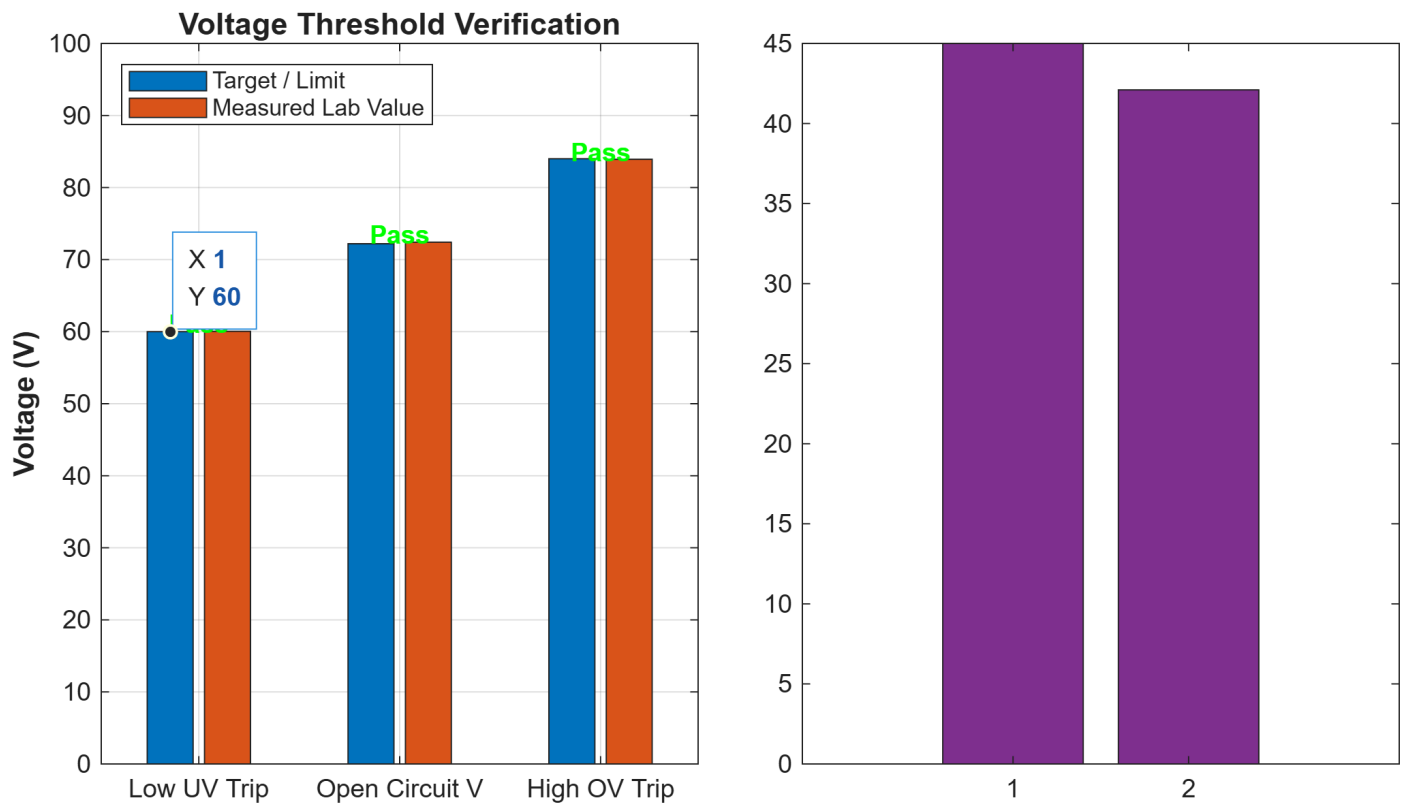
% --- Subplot 2: Thermal Metric ---
nexttile;
thermal_label = {'Peak Thermal Rise'};
thermal_ceiling = 45.0;
thermal_measured = 42.1;

b2 = bar([thermal_ceiling; thermal_measured]', 'grouped');
b2(1).FaceColor = '#7E2F8E'; % Ceiling Purple

nexttile(1)
TargetLimitBar = findobj(gcf, "DisplayName", "Target / Limit");

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datatip(TargetLimitBar,0.8571,60);
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b2(2).FaceColor = '#A2142F'; % Measured Red
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Unrecognized property 'FaceColor' for class 'matlab.graphics.GraphicsPlaceholder'.

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grid on;
xticklabels(thermal_label);
ylabel('Temperature (\circC)', 'FontSize', 11, 'FontWeight', 'bold');
title('Thermal Boundary Verification', 'FontSize', 12, 'FontWeight', 'bold');
legend({'Absolute Ceiling', 'Stabilized Lab Value'}, 'Location',
'northwest');
text(1, thermal_measured+1, 'Pass (Safe)', 'HorizontalAlignment', 'center',
'FontWeight', 'bold', 'Color', 'g');
ylim([0, 55]);

%% =====
%% FIGURE 2: CHEMISTRY COMPARISON (NMC vs LFP)
%% =====
figure('Name', 'Chemistry Comparison: NMC vs LFP', 'Color', 'w', 'Position',
[150, 150, 1000, 600]);
t = tiledlayout(2, 2, 'TileSpacing', 'Detailed', 'Padding', 'Compact');
title(t, 'Battery Chemistry Performance Comparison', 'FontSize', 14,
'FontWeight', 'bold');

chem_names = {'NMC', 'LFP'};
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% --- 1. Nominal Cell Voltage ---
nexttile;
% Format: [NMC_mean, LFP_mean]
v_means = [3.65, 3.2];
v_err    = [0.05, 0.0]; % Range bounds
b = bar(chem_names, v_means, 'FaceColor', 'flat');
b.CData(1,:) = [0 0.4470 0.7410]; b.CData(2,:) = [0.4660 0.6740 0.1880];
hold on;
errorbar(1:2, v_means, v_err, 'k', 'LineStyle', 'none', 'LineWidth', 1.5);
grid on; ylabel('Voltage (V)', 'FontWeight', 'bold');
title('Nominal Cell Voltage');
ylim([0, 4.5]);

% --- 2. Specific Gravimetric Density ---
nexttile;
d_means = [185, 105];
d_err    = [35, 15];
b = bar(chem_names, d_means, 'FaceColor', 'flat');
b.CData(1,:) = [0 0.4470 0.7410]; b.CData(2,:) = [0.4660 0.6740 0.1880];
hold on;
errorbar(1:2, d_means, d_err, 'k', 'LineStyle', 'none', 'LineWidth', 1.5);
grid on; ylabel('Specific Energy (Wh/kg)', 'FontWeight', 'bold');
title('Specific Gravimetric Density');
ylim([0, 260]);

% --- 3. Thermal Runaway Threshold ---
nexttile;
t_runaway = [210, 270];
b = bar(chem_names, t_runaway, 'FaceColor', 'flat');
b.CData(1,:) = [0 0.4470 0.7410]; b.CData(2,:) = [0.4660 0.6740 0.1880];
grid on; ylabel('Temperature (\circC)', 'FontWeight', 'bold');
title('Thermal Runaway Threshold');
ylim([0, 320]);

% --- 4. Cycle Longevity (80% DoD) ---
nexttile;
c_means = [1500, 3500];
c_err    = [500, 1500];
b = bar(chem_names, c_means, 'FaceColor', 'flat');
b.CData(1,:) = [0 0.4470 0.7410]; b.CData(2,:) = [0.4660 0.6740 0.1880];
hold on;
errorbar(1:2, c_means, c_err, 'k', 'LineStyle', 'none', 'LineWidth', 1.5);
grid on; ylabel('Cycles to 80% EOL', 'FontWeight', 'bold');
title('Cycle Longevity (80% DoD)');
ylim([0, 5500]);

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